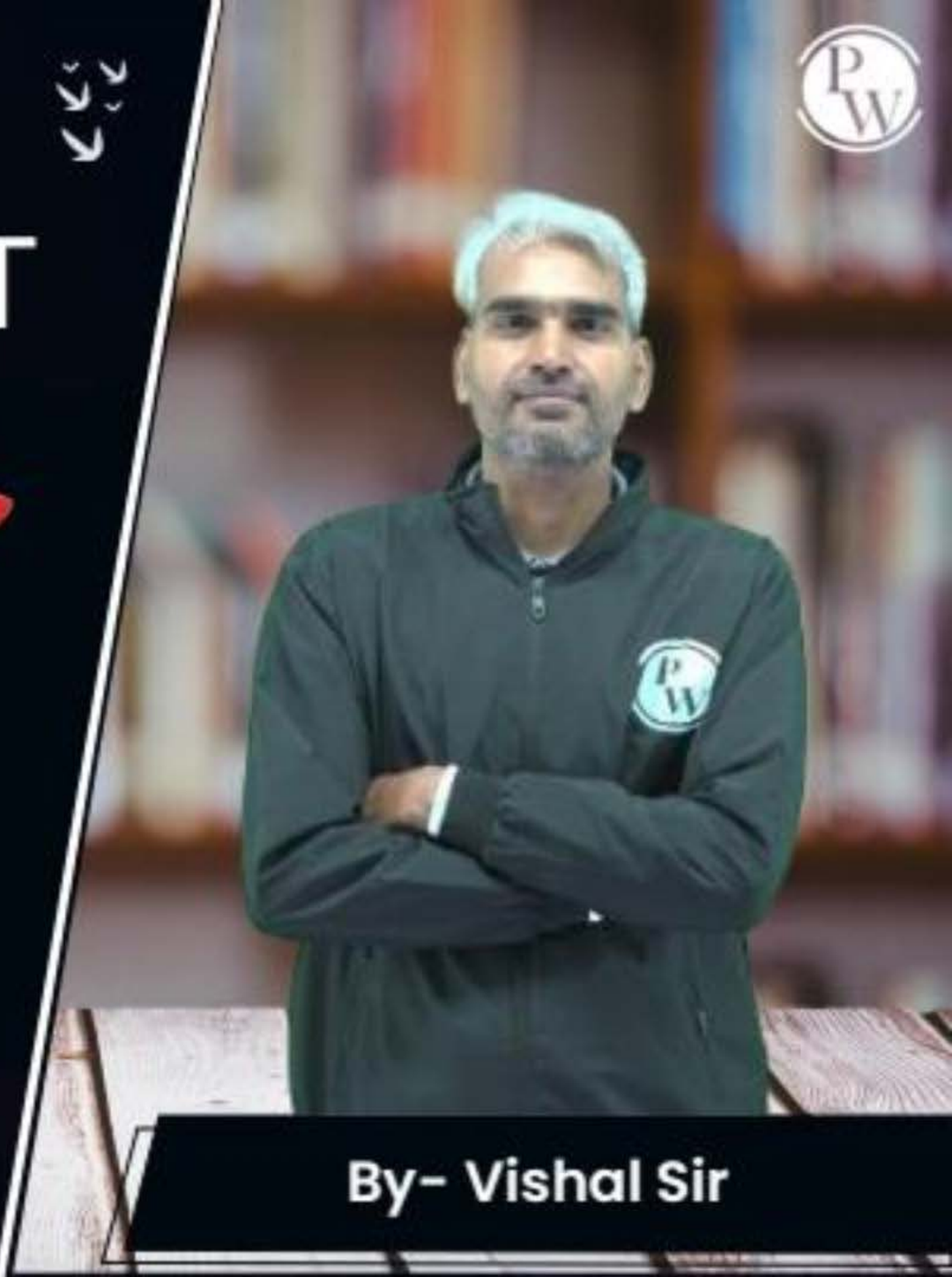
# Computer Science & IT



## Database Management System

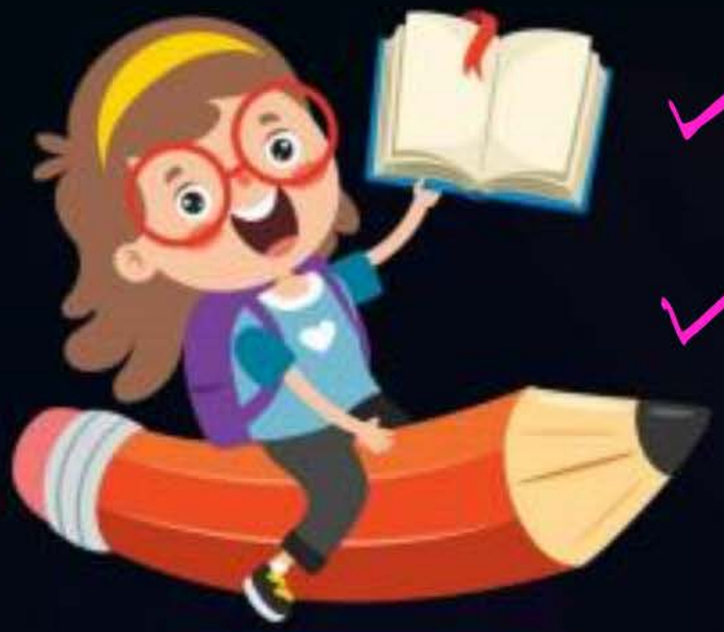
## Relational Model & Normal Forms

Lecture No. 03



By- Vishal Sir

# Recap of Previous Lecture



- Topic: Limitations of file system
- Topic: Introduction Relational Database Model
- Topic: Functional dependency

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# Topics to be Covered



Topic Functional dependency

Topic Types of functional dependency

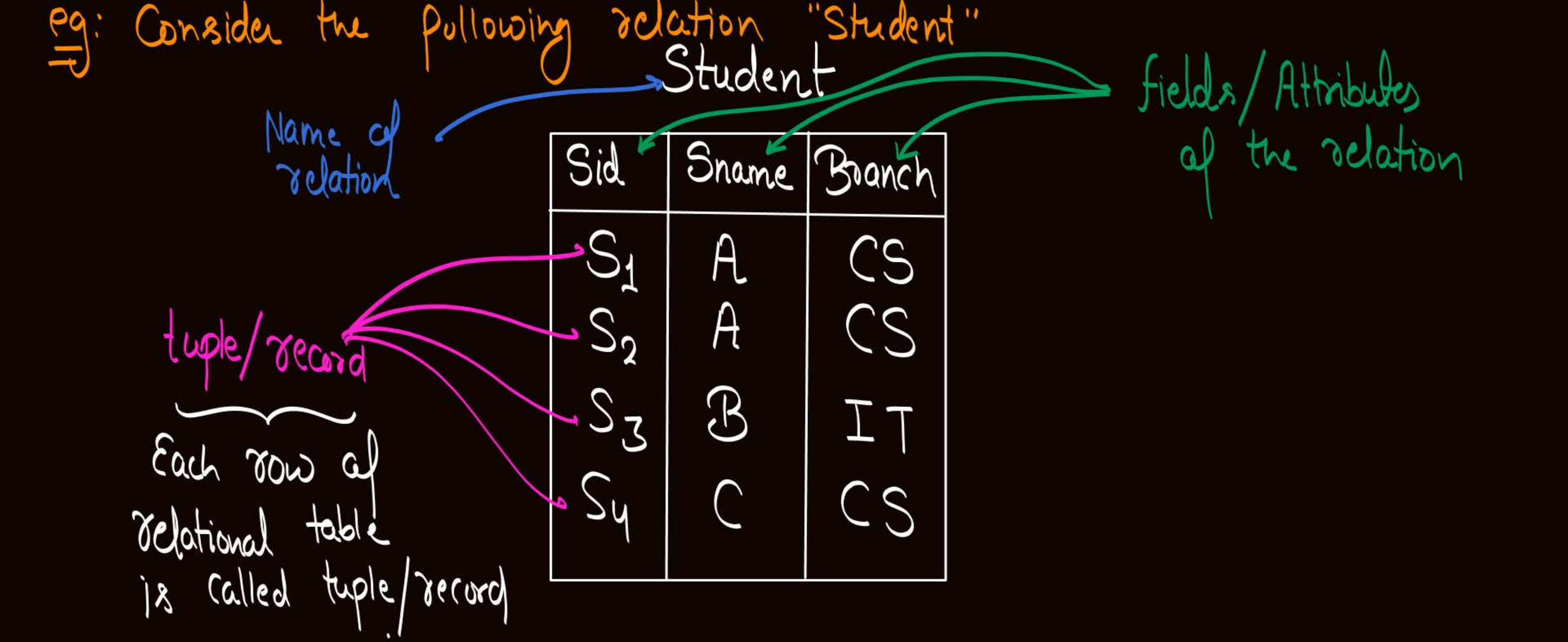
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# Topic : Relational database

- In relational database information will be organized in the form of table {i.e:, Collection of rows and Columns}
- Each table of relational database is called a relation
- Dr. Codd defined 13 rules {from "rule 0" to "rule 12"} for a table to be called a relation.

eg: Consider the following relation "Student"



* **Degree / Arity :** The number of attributes (Columns) in the relation is defined as degree/arity of that relation

* **Cardinality :-** Number of tuples/record in a relational table define the cardinality of that relation

* **Relational Schema :-** Relational schema provides the abstract detail of the relation

eg: Name_of_Relation( 1st attribute in Relⁿ , 2nd attribute in Relⁿ , 3rd attribute in Relⁿ , ---- )

order is important

* **Relational instance :-** If the data/record are present in the relation, then Set of all those tuple will define the relational instance of that time

{ relational instance may change with time because of insert, delete or update oph }

eg: Consider the following relation "Student"

Student

Name of relation

fields / Attributes of the relation

| Sid | Sname | Branch |
|---|---|---|
| $S_1$ | A | CS |
| $S_2$ | A | CS |
| $S_3$ | B | IT |
| $S_4$ | C | CS |

* Degree/Arity = 3
* Cardinality = 4
* Relational Schema w.r.t. Student Relation is
  Student (Sid, Sname, Branch)
* Relational instance (for current instance) = { $(S_1, A, CS)$, $(S_2, A, CS)$, $(S_3, B, IT)$, $(S_4, C, CS)$ }



# Topic : Functional dependency (FD)

Functional dependencies are used to define the relationship among the attributes of relation

Functional dependencies are denoted by format,

$$X \longrightarrow Y$$

Where X and Y are some set of attributes from the relation



# Topic : Functional dependency (FD)

In functional dependency $X \longrightarrow Y$,

'X' is called determinant,

& 'Y' is called dependent.



# Topic : Functional dependency (FD)

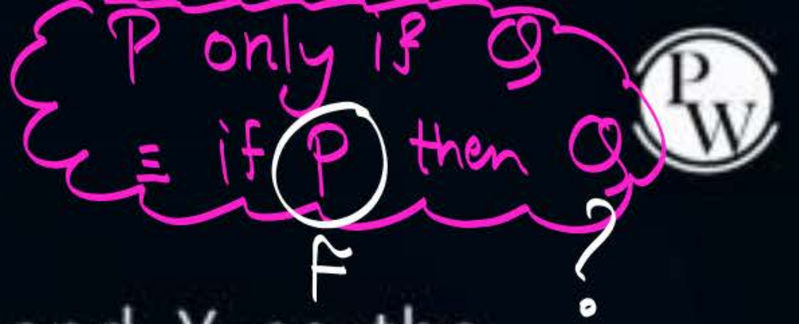
- Functional dependency defines the relationship between two sets of attributes in a relational table.

- It states that the values of dependent attribute set can be completely determined by the values of determinant attributes set. In other words, if a determinant attribute's value is known, the value of the dependent attribute can be deduced.

i.e. in $X \longrightarrow Y$
if we know the value of X
then we can deduce the value of Y from the relation



# Topic : Functional dependency (FD)




- Let R be the relational schema with X and Y as the attribute sets over relation R.

(Functional dependency X→Y exists in R) only if
(For all pair of tuples t1,t2∈R
If t1.X = t2.X then t1.Y = t2.Y)

* if Functional dependency X→Y exists in R
then
(for ∀ $t_1, t_2 \in R$
if $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$)



# Topic : Functional dependency (FD)

if $P \rightarrow Q$
then $Q \rightarrow P$

* If functional dependency $X \rightarrow Y$ exists in relation $R$, then

Whenever values w.r.t. attribute set X are same in any pair of tuples, then values of attribute set Y will also be same in those tuples

It is the necessary condition for $X \rightarrow Y$ to exist in the relation, but not sufficient w.r.t. relational instance

But if values w.r.t attribute set Y are same in any pair of tuples, then values w.r.t. attribute set X need not be same in those tuples.

* Necessary Cond$^n$ for $X \longrightarrow Y$ to exist in relation R is,
Whenever values w.r.t. attributes of set X are same in any pair of tuples, then values w.r.t. attributes of set Y should also be same in those tuples.

* If there exists two tuples in which values of attributes of set X are same, but values w.r.t. attributes of set Y are not same then functional dependency $X \longrightarrow Y$ can never exist in that relation.



# Topic : Functional dependency (FD)

- If necessary condition for functional dependency "X→Y" does not hold true based on given relation instance, then functional dependency "X→Y" can never exist in the given relation.

- Even if necessary condition for functional dependency "X→Y" does hold true based on given relation instance, then also we can not be sure whether functional dependency X→Y exists in the relation or not, because it is just the relational instance.

eg: Consider the following relational instance

| Sid | Sname | Branch |
|---|---|---|
| 1 | Ram | CS |
| 2 | Ram | CS |
| 3 | John | IT |
| 4 | Mohan | CS |
| 5 | Rohit | IT |

Whenever values of Sid are same Values of Sname are also same → All values of Sid are unique, i.e there are no tuples in which Sids are same, hence we don't need to worry about Sname

And Necessary condⁿ for functional dependency Sid → Sname is satisfied

Hence, Sid → Sname may hold in the relation.

eg: Consider the following relational instance

| Sid | Sname | Branch |
|---|---|---|
| 1 | Ram | CS |
| 2 | Ram | CS |
| 3 | John | IT |
| 4 | Mohan | CS |
| 5 | Rohit | IT |
| 6 | Ram | IT |

For every pair of tuples in relational instance whenever values of Sname are same, values of Branch are also same in those tuples.

i.e, Necessary condn for functional dependency Sname → Branch is satisfied

Hence, Sname → Branch may hold in relation

But we can not conclude that Sname → Branch holds in the relⁿ

eg: Consider the following relational instance

| Sid | Sname | Branch |
|---|---|---|
| 1 | Ram | CS |
| 2 | Ram | CS |
| 3 | John | IT |
| 4 | Mohan | CS |
| 5 | Rohit | IT |

→ There exists some tuples in the relation in which values w.r.t. "Branch" are same but values w.r.t. "Sname" are not same

i.e., necessary condition for

Branch ⟶ Sname does not hold in the relation based on relational instance,

↳ Hence, we can conclude that

Branch ⟶ Sname will never hold in this relation.

#Q. From the following instance of a relation schema R (A,B,C), we can conclude that:

| A | B | C |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 0 |
| 2 | 3 | 2 |
| 2 | 3 | 2 |

We can conclude that B does not functionally determines C { Because necessary Condⁿ for B → C is dis-satisfied

(A) A functionally determines B, and B functionally determines C — it may be true, it may not be true; it is wrong

(B) A functionally determines B, and B does not functionally determine C — it is dairy; it is true

(C) B does not functionally determine C — it is true

(D) A does not functionally determine B, and B does not functionally determine C — it is true; We can not conclude this

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 2 | 3 |
| 1 | 2 | 4 |
| 2 | 2 | 1 |
| 3 | 1 | 2 |
| 4 | 1 | 2 |

$A \not\rightarrow C$

$A \rightarrow B$

$AB \rightarrow C$

$BC \rightarrow A$

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $A \rightarrow C$

b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 2 | 3 |
| 1 | 2 | 4 |
| 2 | 2 | 1 |
| 3 | 1 | 2 |
| 4 | 1 | 2 |

$A \not\rightarrow C$

$A \rightarrow B$ may hold true

$AB \rightarrow C$

$BC \rightarrow A$

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $A \rightarrow C$

b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 2 | 3 |
| 1 | 2 | 4 |
| 2 | 2 | 1 |
| 3 | 1 | 2 |
| 4 | 1 | 2 |

$A \not\rightarrow C$

$A \rightarrow B$ may hold

$AB \not\rightarrow C$

$BC \rightarrow A$

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $A \rightarrow C$

b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 2 | 3 |
| 1 | 2 | 4 |
| 2 | 2 | 1 |
| 3 | 1 | 2 |
| 4 | 1 | 2 |

$A \not\rightarrow C$

$A \rightarrow B$ may hold

$AB \not\rightarrow C$

$BC \not\rightarrow A$

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $A \rightarrow C$

b) $A \rightarrow B$

c) $AB \rightarrow C$

d) $BC \rightarrow A$

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 2 | 2 |
| 2 | 4 | 3 |
| 3 | 3 | 4 |
| 4 | 1 | 5 |
| 5 | 3 | 6 |

All values of C are unique

Which of the following functional dependencies may hold true (not necessarily) based on given relational instance.

a) $B \not\rightarrow C$

b) $A \not\rightarrow B$

c) $C \rightarrow B$

d) $B \not\rightarrow A$

e) $C \rightarrow A$

Note:

Functional dependencies that holds in a relation are identified by database designer, and mostly functional dependencies are given in the question



# Topic : Types of Functional Dependency

There are two types of Functional dependencies

- Trivial Functional dependency
- Non-trivial functional dependency

We can define a third type of functional dependency
i.e. Semi-non-trivial functional dependency.
↳ it is a mixture of trivial and non-trivial FD



# Topic : Trivial Functional Dependency

- Let X and Y are two set of attributes from relation R.

Functional dependency $X \rightarrow Y$ is a trivial functional dependency if and only if $X \supseteq Y$

Eg: Sid $\rightarrow$ Sid
Sid, Sname $\rightarrow$ Sid
Sid, Sname $\rightarrow$ Sname

} All are trivial functional dependency

+ Every trivial functional dependency possible with the attributes of relation R will always hold true in the relation R



# Topic : Non-trivial Functional Dependency

- Functional dependency $X \rightarrow Y$ is called a non-trivial functional dependency only if,

$$X \cap Y = \phi$$

Where X & Y are any two non-Empty sets.

* All possible non-trivial functional dependencies that can be defined using the attributes of relation R, need not hold true in the relation R.

eg: Consider the following relational schema
Student (Sid, Sname, Branch)
Write all possible non-trivial functional dependencies w.r.t. attributes of relation "Student"

Sol":

Sid → Sname
Sid → Branch
Sname → Sid
Sname → Branch
Branch → Sid
Branch → Sname
Sid → Sname, Branch
Sname → Sid, Branch
Branch → Sid, Sname
Sid, Sname → Branch
Sid, Branch → Sname
Sname, Branch → Sid



# Topic : Semi non-trivial Functional Dependency

- Functional dependency $X \longrightarrow Y$ is called a semi non-trivial functional dependency only if,

  (i) $X \not\supseteq Y$

  and (ii) $X \cap Y \neq \phi$

Eg: R(Sid, Sname, Branch)

Sid, Sname $\longrightarrow$ Sname, Branch

Sname: trivial Part

Branch: Non-trivial Part

$\{Sid, Sname\} \not\supseteq \{Sname, Branch\}$

and

$\{Sid, Sname\} \cap \{Sname, Branch\} = \{Sname\} \neq \phi$

$\Rightarrow$ Semi non-trivial FD

* We will be able to split semi non-trivial FD into two parts, i.e., trivial part and non-trivial part,
  so We don't need to discuss the semi non-trivial FDs seperately.

All Possible non-trivial FDs w.r.t. three attributes A, B & C

#Q. Consider the following relational instance

| A | B | C |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 2 |
| 2 | 1 | 2 |
| 2 | 2 | 3 |
| 3 | 3 | 4 |

AC | B
(1,1) | 1
(1,2) | 1
(2,2) | 1
(2,3) | 2
(3,4) | 3

$A \rightarrow B$ ✗ $A \rightarrow C$ ✗ $B \rightarrow A$ ✗ $B \rightarrow C$ ✗ $C \rightarrow A$ ✗ $C \rightarrow B$ ✓

$A \rightarrow BC$ ✗ $B \rightarrow AC$ ✗ $C \rightarrow AB$ ✗

$AB \rightarrow C$ ✗ $BC \rightarrow A$ ✗ $AC \rightarrow B$ ✓

Find all non-trivial FDs which may hold true in the above relation based on given relational instance.

Ans = $C \rightarrow B$ & $AC \rightarrow B$

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H.W:

FD $X \rightarrow Y$ is called an useful FD if and only if both $X$ and $Y$ are non-Empty sets and

$$X \cap Y = \phi$$

How many useful FDs are possible in a relation with '4' attributes ?

H.W.

Q:- How many non-trivial FDs are possible in a relation with "4" attributes



# 2 mins Summary

t.me/vishalRawtiyapw

**Topic** Functional dependency

**Topic** Types of functional dependency

Slide

# THANK - YOU